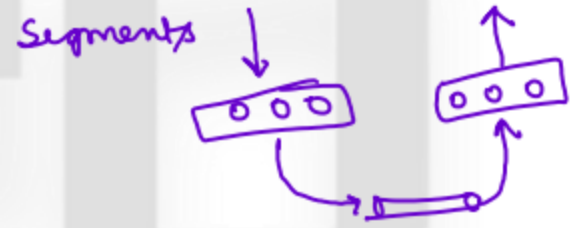
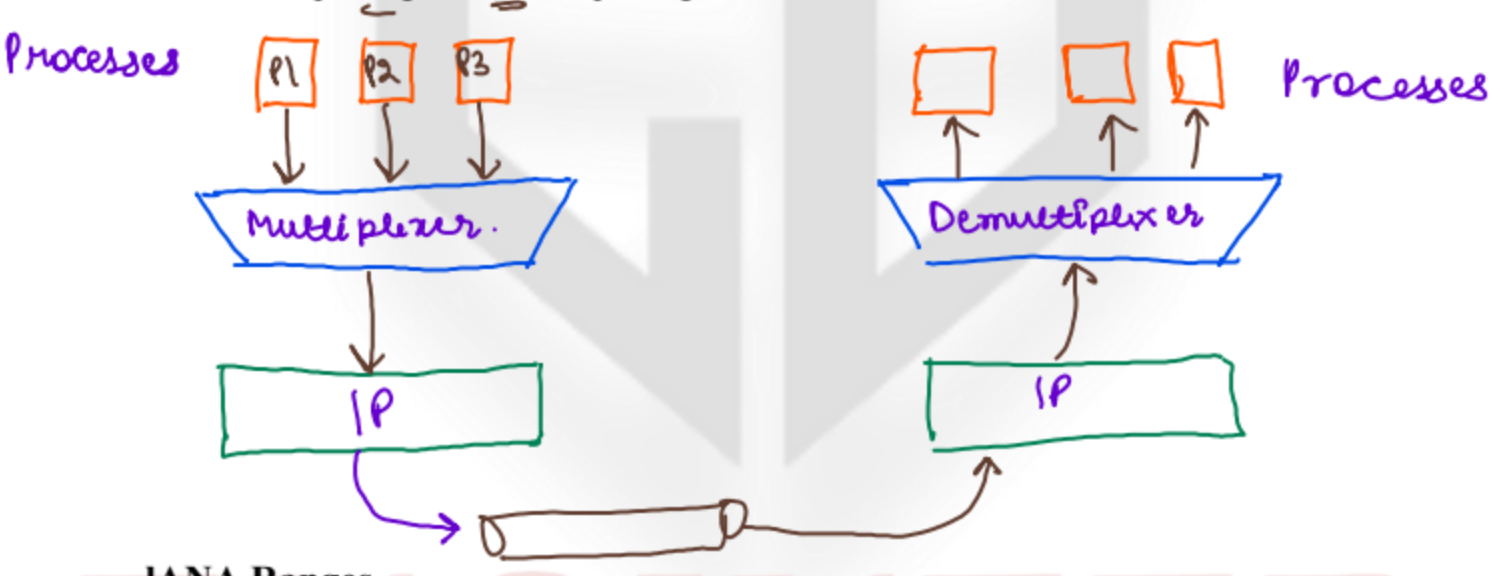
Transport Layer: Process-to-process delivery, Transport layer protocols (UDP and TCP), Multiplexing, Connection management, Flow control and retransmission, Window management, TCP Congestion control, Quality of service.

Services of Transport Layer



- Segmentation and Reassembly
- Process to process delivery
- Use port addressing.
- TCP
- UDP
- Multiplexing and Demultiplexing



lANA Ranges

•The lANA (Internet Assigned Number Authority) has divided the port numbers into three ranges: well known, registered, and dynamic (or private).

- Well-known ports: The ports ranging from 0 to 1023 are assigned and controlled by lANA. These are the well-known ports.

20, 21
FTP
HTTP
SMTP = 25

- Registered ports: The ports ranging from 1024 to 49,151 are not assigned or controlled by lANA. They can only be registered with lANA to prevent duplication.

- Dynamic ports: The ports ranging from 49,152 to 65,535 are neither controlled nor registered. They can be used by any process. These are the ephemeral ports.

Port no.	Protocol
23	TELNET (Teletype Network Protocol)
25	SMTP (Simple Mail Transfer protocol)
53	DNS (Domain Name server)
80	HTTP (Hypertext Transfer Protocol
111	RPC (Remote Procedure Call)
20	FTP, Data connection
21	FTP, control connection

Socket Addresses

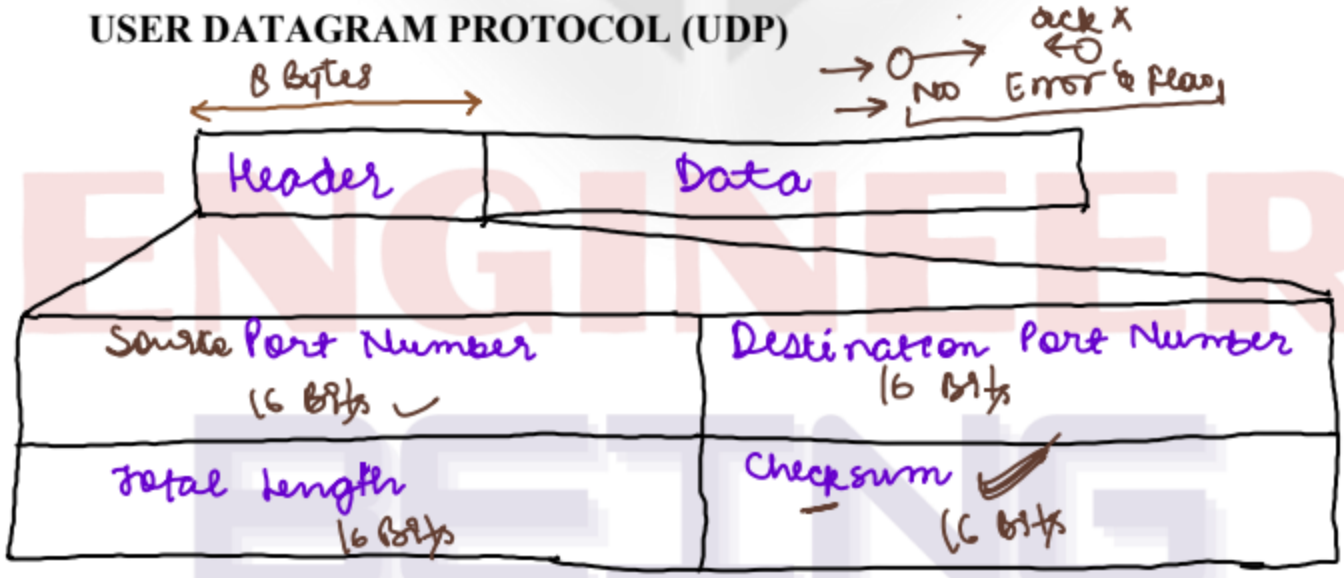
The combination of an **IP address and a port number** is called a socket address.

Ques : If the data link layer is reliable and has flow and error control, do we need this at the transport layer, too?

The answer is yes.

- Reliability at the data link layer is between two nodes; we need reliability between two ends.
- Because the network layer in the Internet is unreliable (best-effort delivery), we need to implement reliability at the transport layer.

USER DATAGRAM PROTOCOL (UDP)

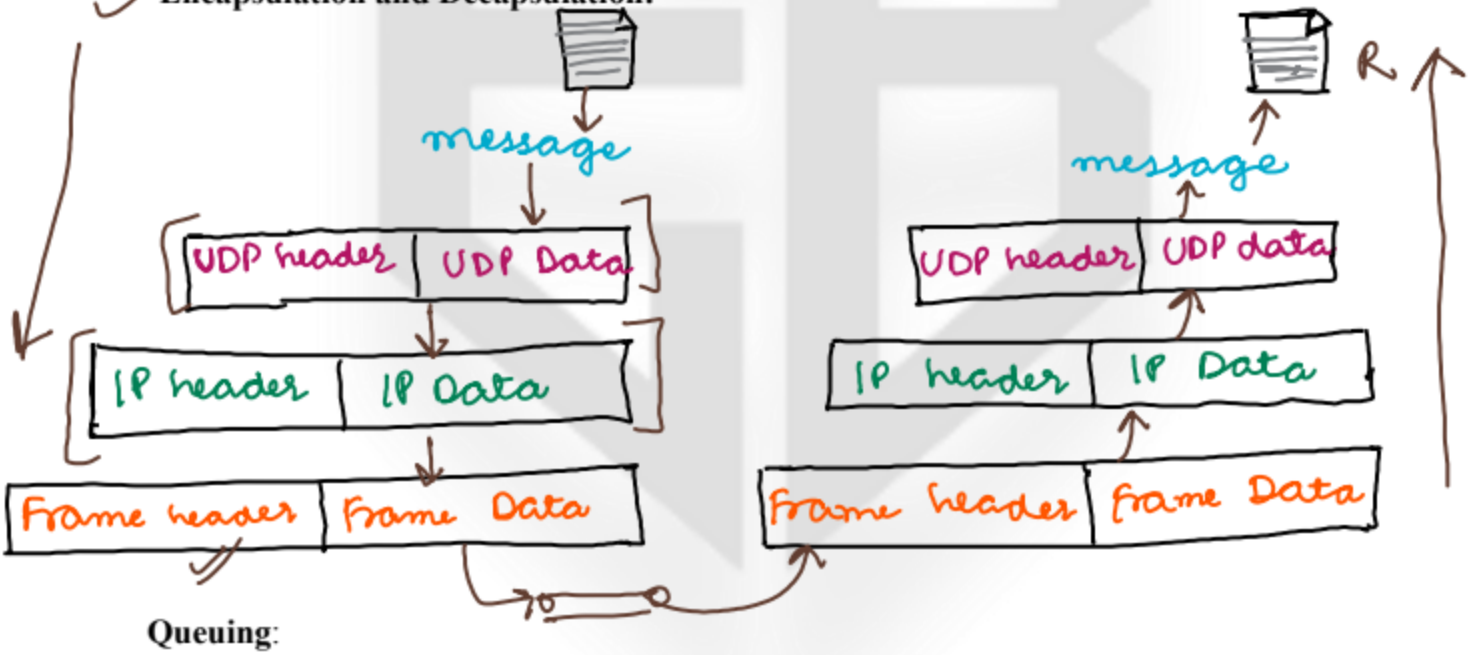


UDP Operations

•**Connectionless Services:** There is no connection establishment and no connection termination, as is the case for TCP. This means that each user datagram can travel on a different path.each request must be small enough to fit into one user datagram. Only those processes sending short messages should use UDP.

•**No Flow and Error Control:** UDP is a very simple, unreliable transport protocol. There is no flow control and hence no window mechanism. The receiver may overflow with incoming messages. There is no error control mechanism in UDP except for the checksum.

•**Encapsulation and Decapsulation:**



Queuing:

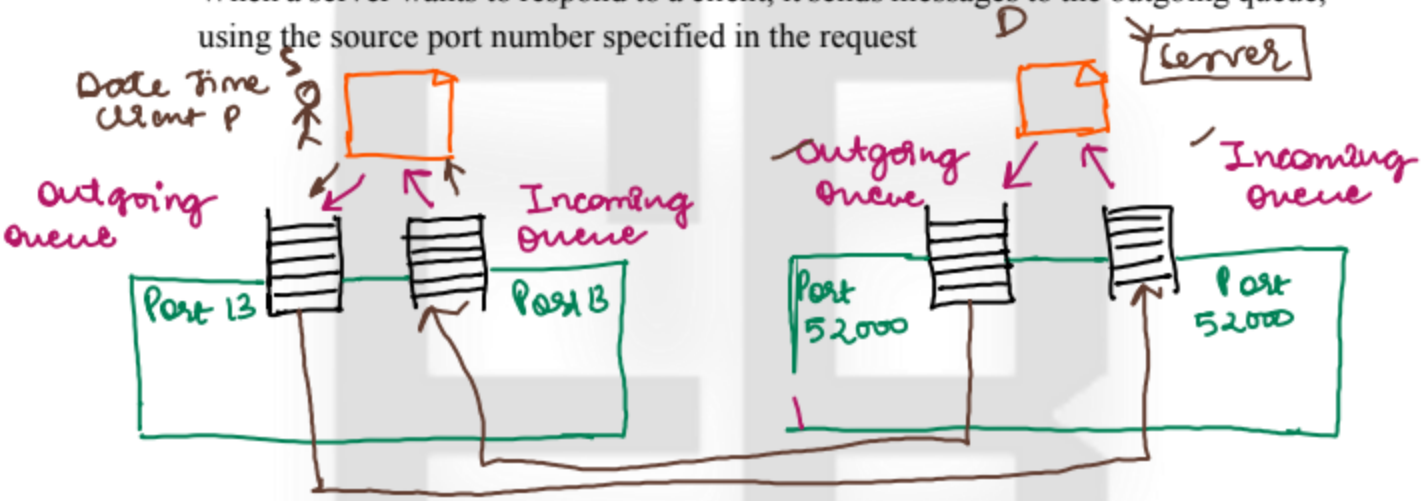
- At the client site, when a process starts, it requests a port number from the operating system.
- The client process can send messages to the outgoing queue by using the source port number specified in the request.
- UDP removes the messages one by one and, after adding the UDP header, delivers them to IP.
- When a message arrives for a client, UDP checks to see if an incoming queue has been created for the port number specified in the destination port number field of the user datagram.
- When a message arrives for a server, UDP checks to see if an incoming queue has been created for the port number specified in the destination port number field of the user datagram.

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- When a server wants to respond to a client, it sends messages to the outgoing queue, using the source port number specified in the request



Uses of UDP

- UDP is suitable for a process that requires simple request-response communication with little concern for flow and error control. It is not usually used for a process such as FTP that needs to send bulk data.
- UDP is suitable for a process with internal flow and error control mechanisms.
- UDP is a suitable transport protocol for multicasting. Multicasting capability is embedded in the UDP software but not in the TCP software.
- UDP is used for management processes such as SNMP (Simple Network Management Protocol).
- UDP is used for some route updating protocols such as Routing Information Protocol.

TCP Services

⇒ TCP Vs UDP (2021, 2018)

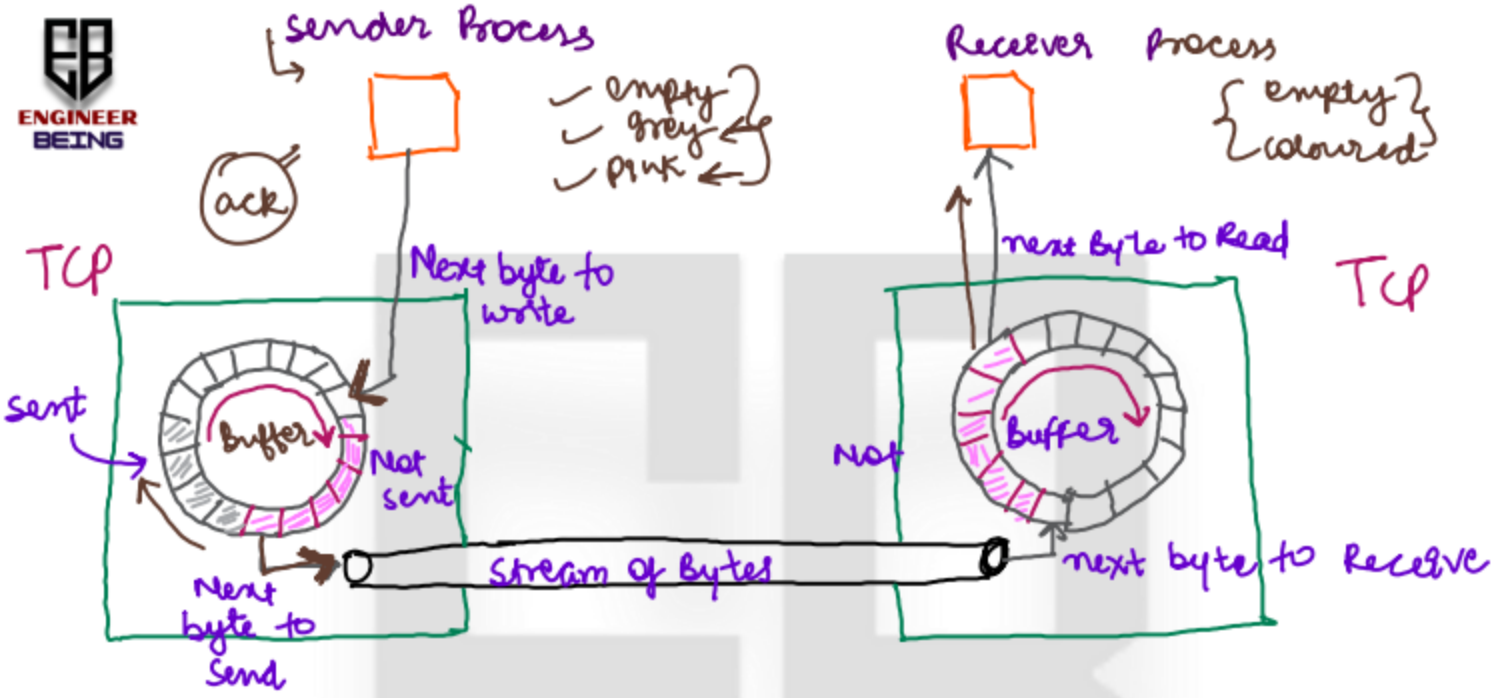
1. **Process-to-Process Communication** (TCP provides process-to-process communication using port number)

Bytes

2. **Stream Delivery Service:** TCP, allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as a stream of bytes. TCP creates an environment in which the two processes seem to be connected by an imaginary "tube" that carries their data across the Internet

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UDP Queue | TCP → Buffer



At the sending site, the buffer has three types of chambers. The white section contains empty chambers that can be filled by the sending process (producer). The gray area holds bytes that have been sent but not yet acknowledged. TCP keeps these bytes in the buffer until it receives an acknowledgment. The colored area contains bytes to be sent by the sending.

At the receiving site, the circular buffer is divided into two areas (shown as white and colored). The white area contains empty chambers to be filled by bytes received from the network. The colored sections contain received bytes that can be read by the receiving process.

2. Full-Duplex Communication



- TCP offers full-duplex service, in which data can flow in both directions at the same time.
- Each TCP then has a sending and receiving buffer, and segments move in both directions.

3. Connection-Oriented Service

TCP, unlike UDP, is a connection-oriented protocol. When a process at site A wants to send and receive data from another process at site B, the following occurs:

- The two TCPs establish a connection between them.
- Data are exchanged in both directions.
- The connection is terminated.



Note that this is a virtual connection, not a physical connection.

4. **Reliable Service:** TCP is a reliable transport protocol. It uses an acknowledgment mechanism to check the safe and sound arrival of data

TCP Features

1. Numbering System

   There are two fields called the sequence number and the acknowledgment number. These two fields refer to the byte number and not the segment number.

2. Byte Number

   The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with a randomly generated number.

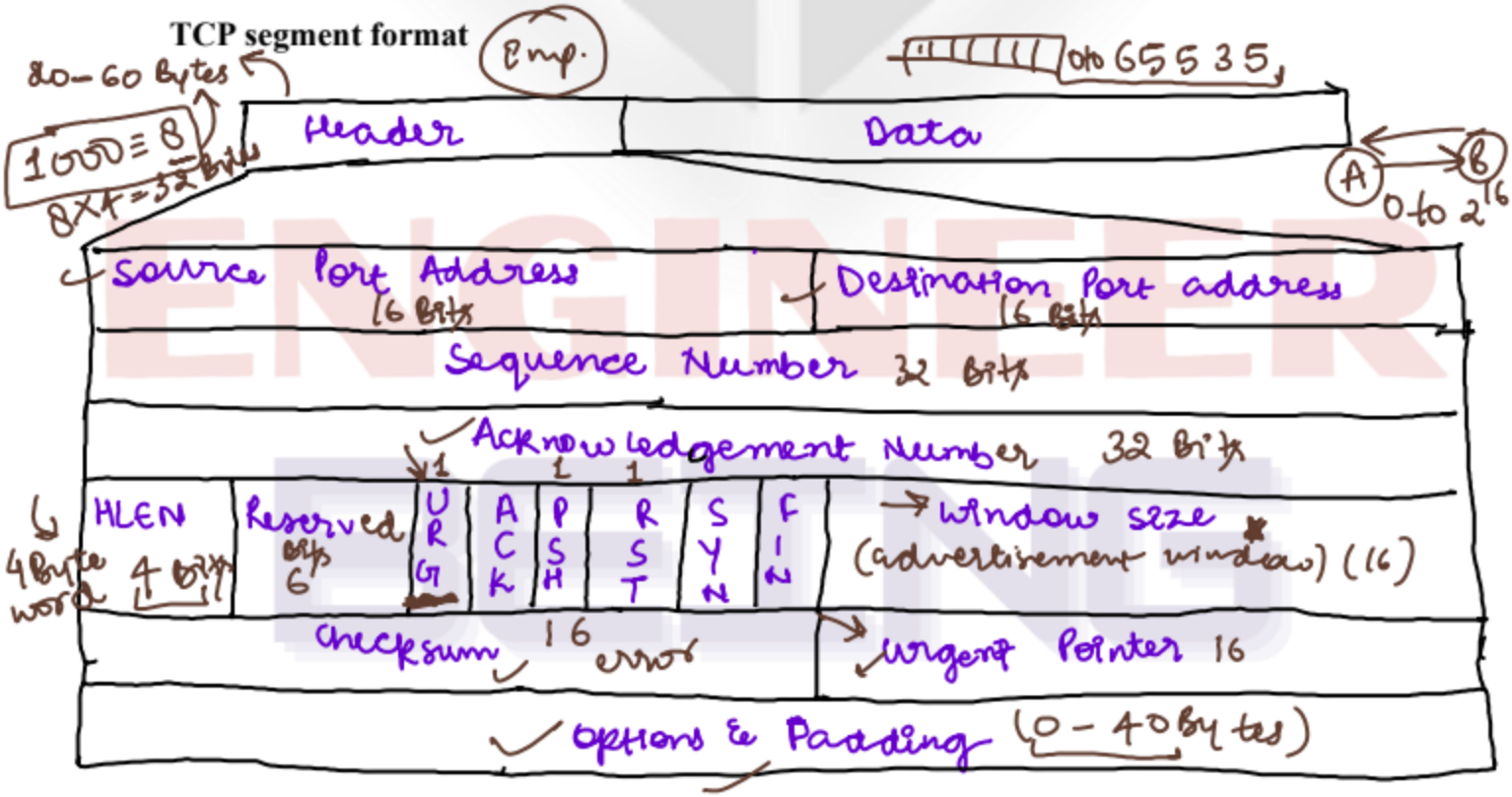
3. Acknowledgment Number

   The value of the acknowledgment field in a segment defines the number of the next byte a party expects to receive.

4. Sequence Number

   After the bytes have been numbered, TCP assigns a sequence number to each segment that is being sent.

TCP segment format



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Source port address: This is a 16-bit field that defines the port number of the the host that is sending the segment.

Destination port address: This is a 16-bit field that defines the port number of the the host that is receiving the segment.

Sequence number: This 32-bit field defines the number assigned to the first byte of data contained in this segment.

Acknowledgment number: This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the other party

Header length: This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes. Therefore, the value of this field can be between 5 (5 x 4 =20) and 15 (15 x 4 =60).

Reserved: This is a 6-bit field reserved for future use.

Control: This field defines 6 different control bits or flags

Flag	Description
URG	The value of urgent pointer field is valid.
ACK	The value of acknowledgement field is valid.
PSH	Push the Data
RST	Reset connection
SYN	Synchronize the sequence no. during connection
FIN	Terminate connection

Window size: This field defines the size of the window, in bytes, that the other party must maintain. Note that the length of this field is 16 bits, which means that the maximum size of the window is 65,535 bytes.

Checksum: This 16-bit field contains the checksum.

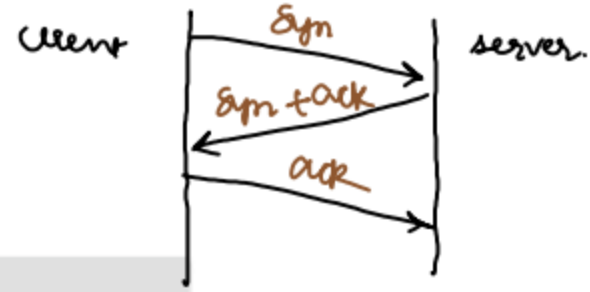
Urgent pointer: This l6-bit field, which is valid only if the urgent flag is set, is used when the segment contains urgent data.

Options: There can be up to 40 bytes of optional information

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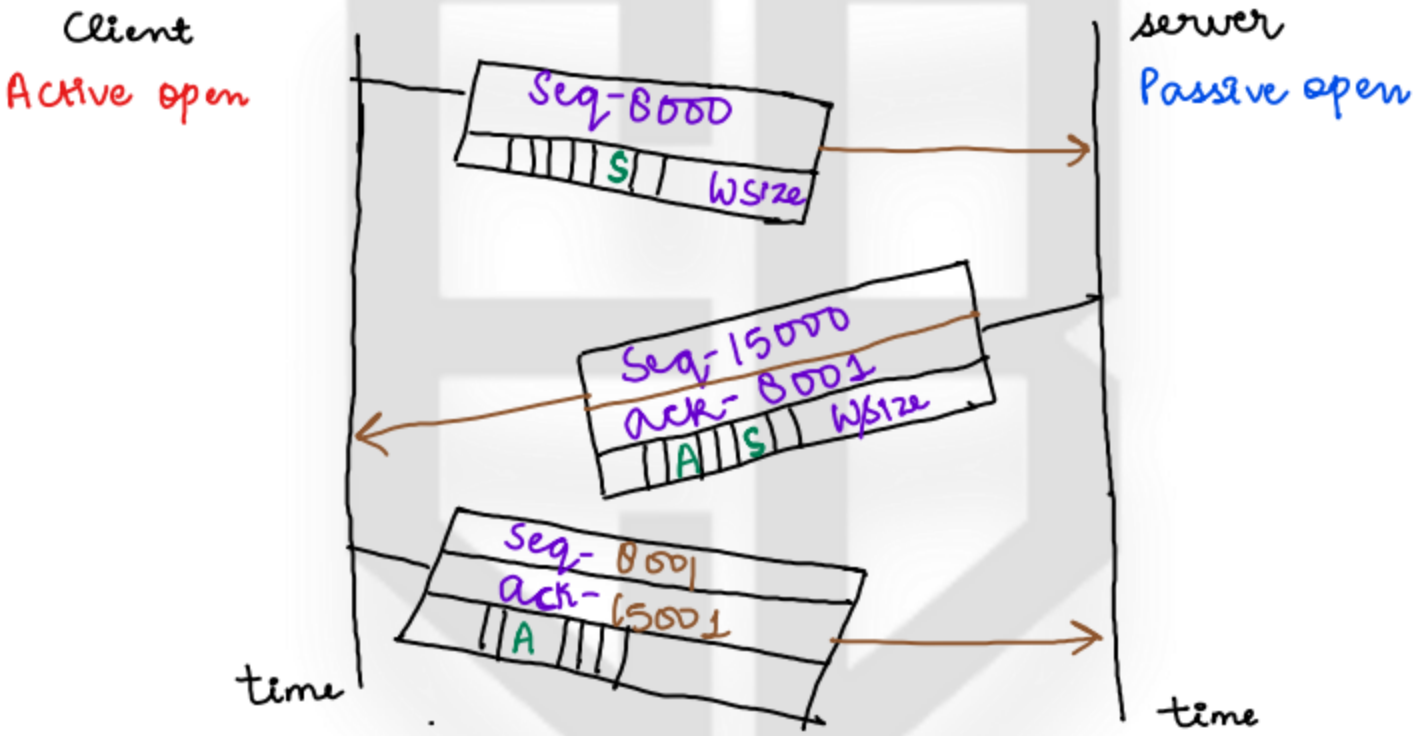
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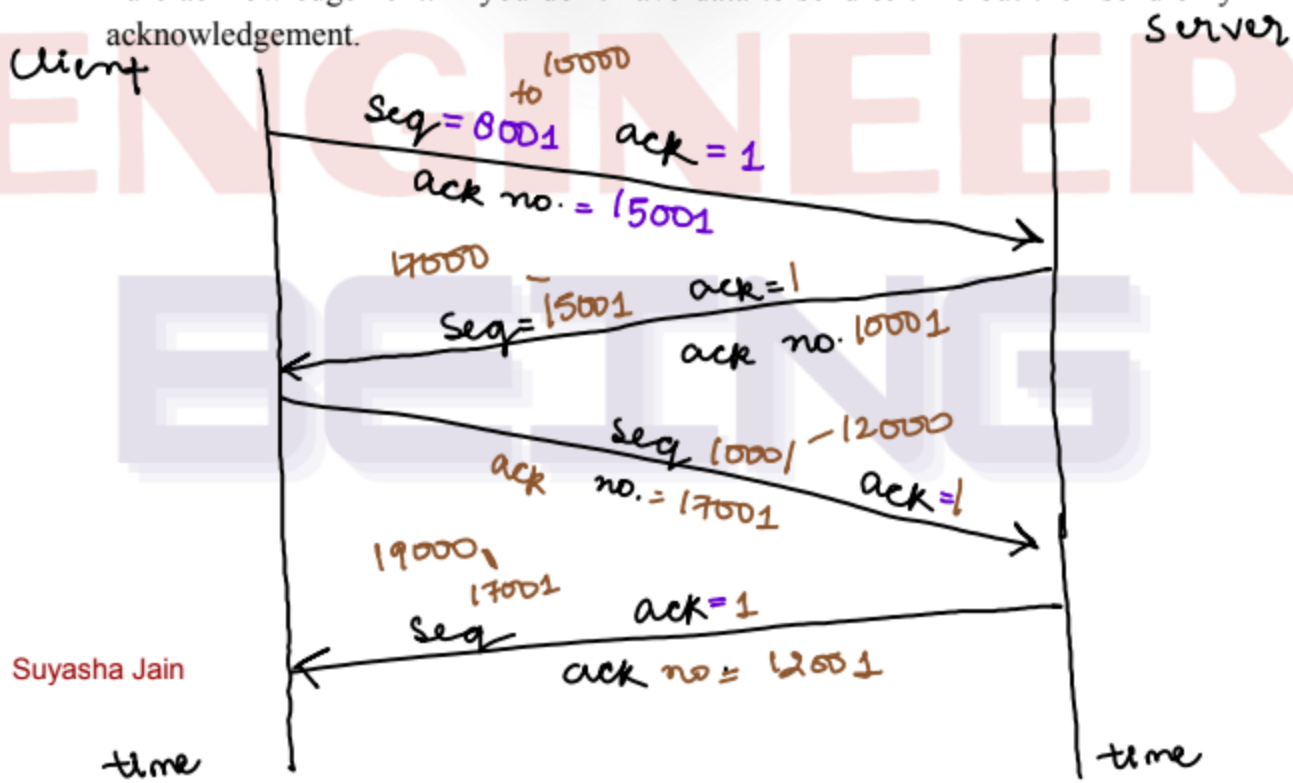
A TCP Connection

Connection Establishment- Three-Way Handshaking (Full duplex mode)



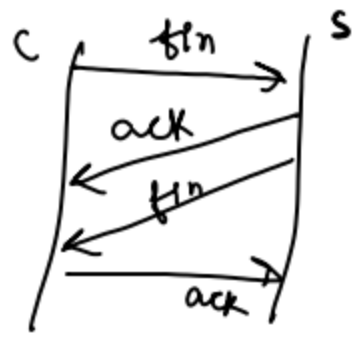
Data Transfer

- Piggybacking - Data and acknowledgement send at same time, this reduces network traffic
- Pure acknowledgement: If you don't have data to send & time out then send only acknowledgement.



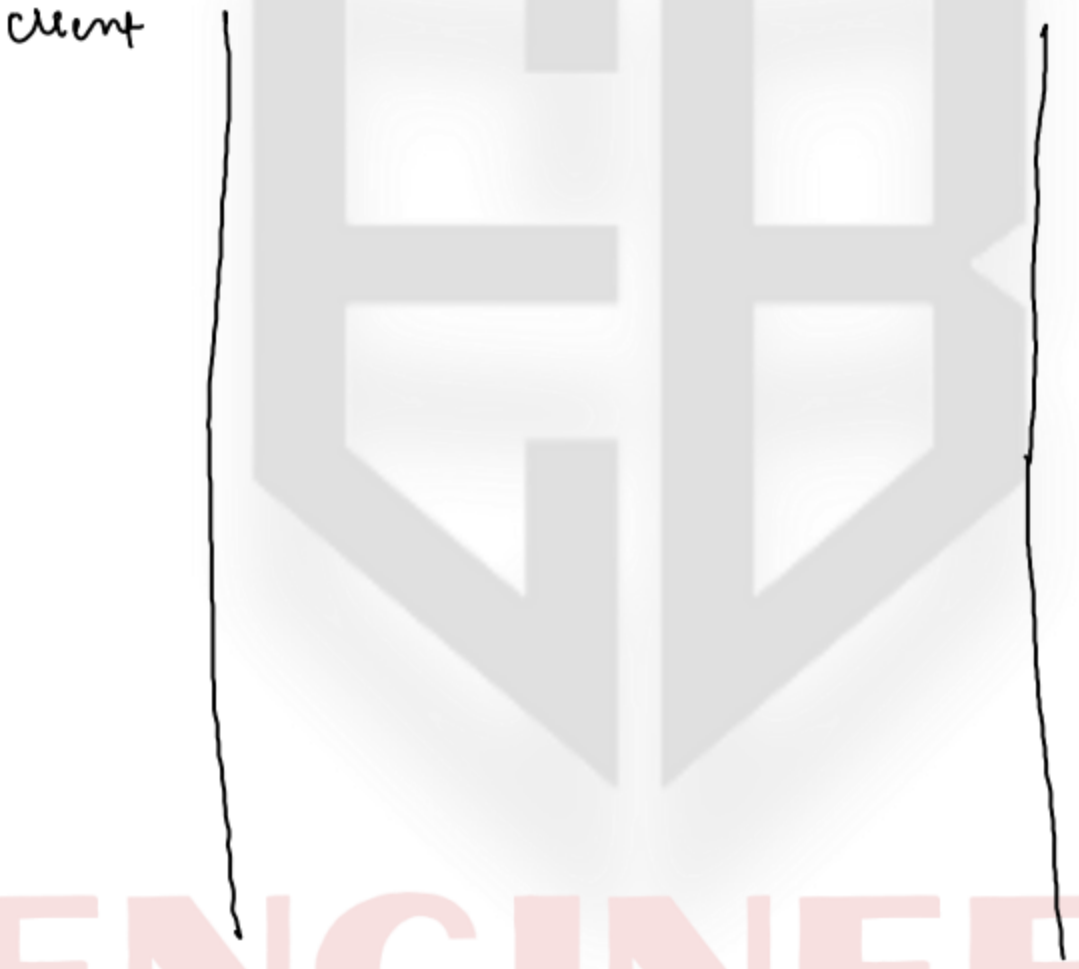
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Connection Termination- when all data is send at client side

Half-Close: client to server connection is over, Data can't be send from client to server, ack can be send



Flow Control in TCP

•TCP, unlike UDP, provides flow control.

•The receiver of the data controls the amount of data that is to be sent by the sender.

•This is done to prevent the receiver from being overwhelmed with data. The numbering system allows TCP to use a byte-oriented flow control.

Error Control

•To provide reliable service, TCP implements an error control mechanism.

•Error detection and correction in TCP is achieved through the use :

•checksum

•acknowledgment

•time-out.

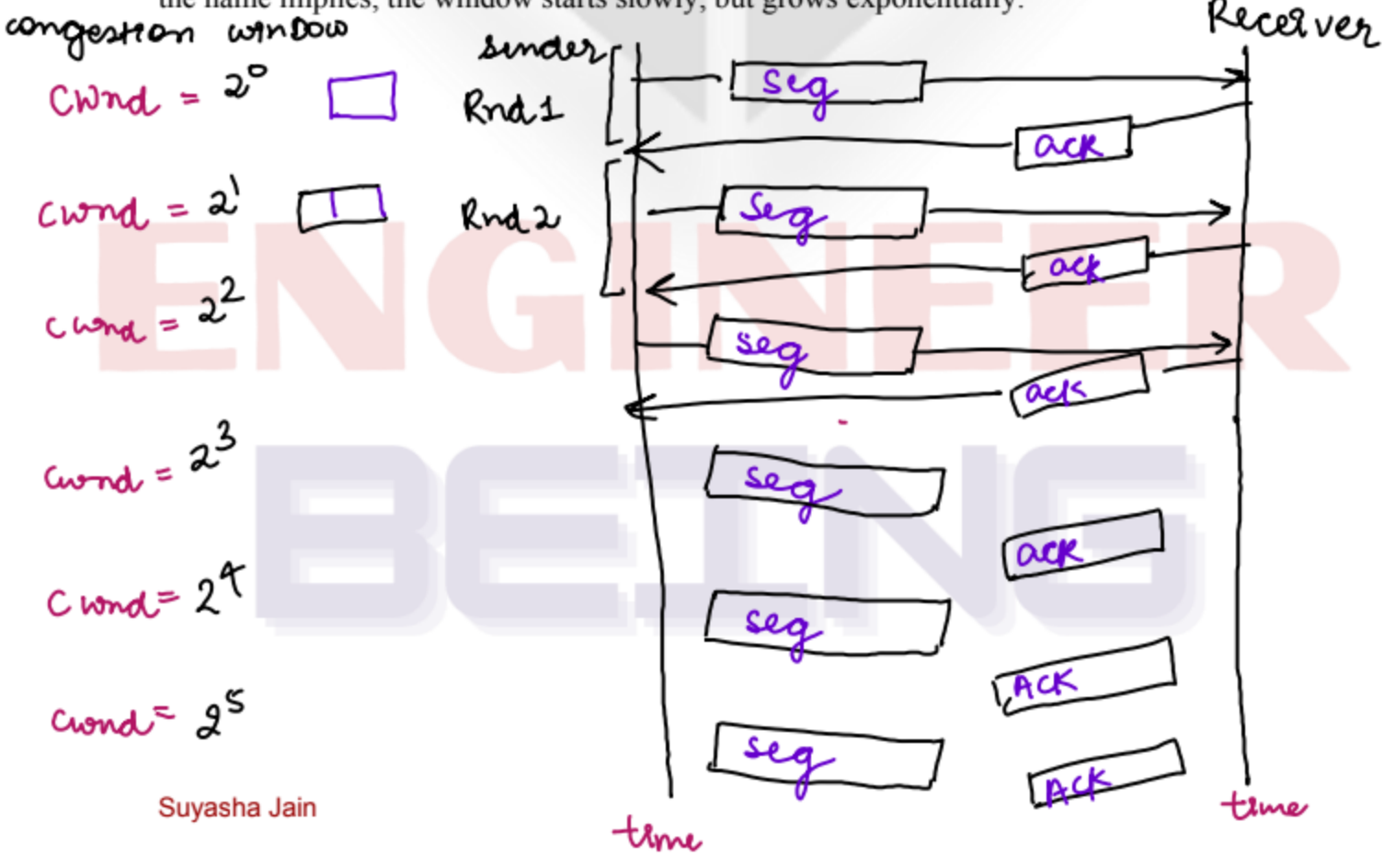
Congestion Control

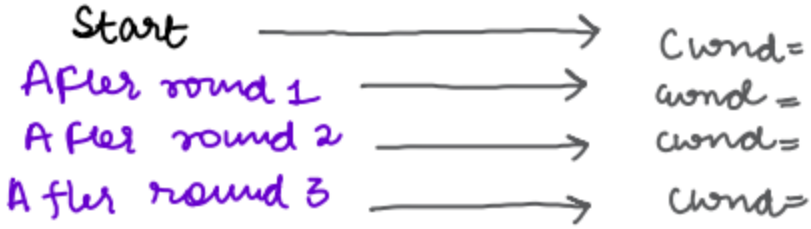
- TCP, unlike UDP, takes into account congestion in the network.
- The amount of data sent by a sender is not only controlled by the receiver (flow control), but is also determined by the level of congestion in the network.

Congestion Policy

1) Slow Start: Exponential Increase

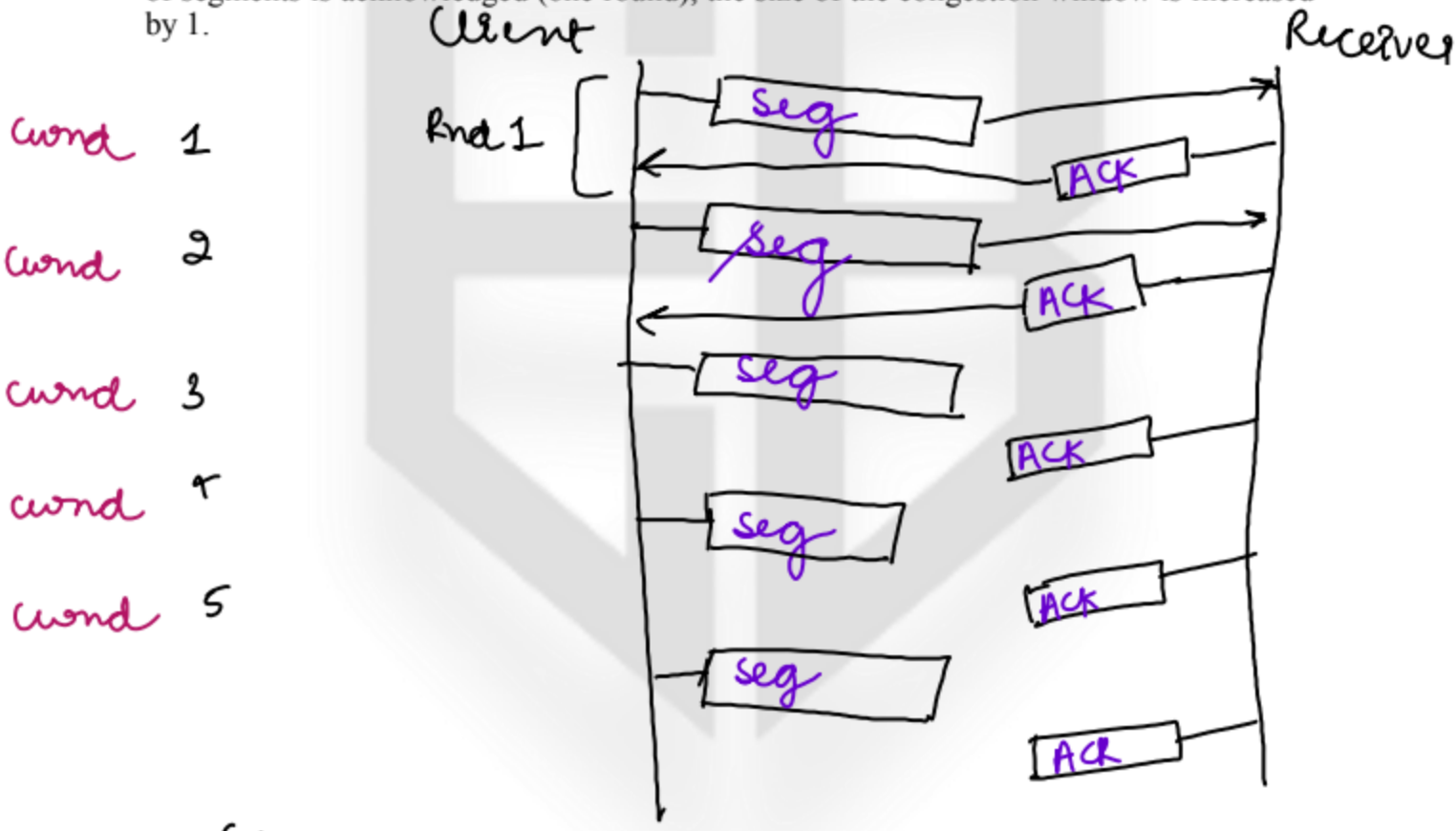
The size of the window increases one MSS each time an acknowledgment is received. As the name implies, the window starts slowly, but grows exponentially.

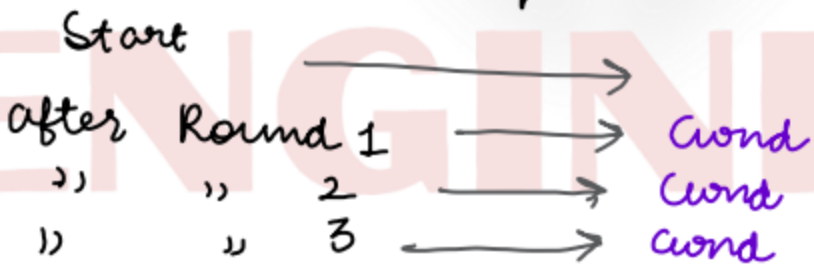




2) Congestion Avoidance: Additive Increase

When the size of the congestion window reaches the slow-start threshold, the slow-start phase stops and the additive phase begins. In this algorithm, each time the whole window of segments is acknowledged (one round), the size of the congestion window is increased by 1.





3) Congestion Detection: Multiplicative Decrease

If congestion occurs, the congestion window size must be decreased.

An implementation reacts to congestion detection in one of the following ways:

o If detection is by time-out, a new slow-start phase starts.

o If detection is by three ACKs, a new congestion avoidance phase starts

QUALITY OF SERVICE

Quality of service (QoS) is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity.

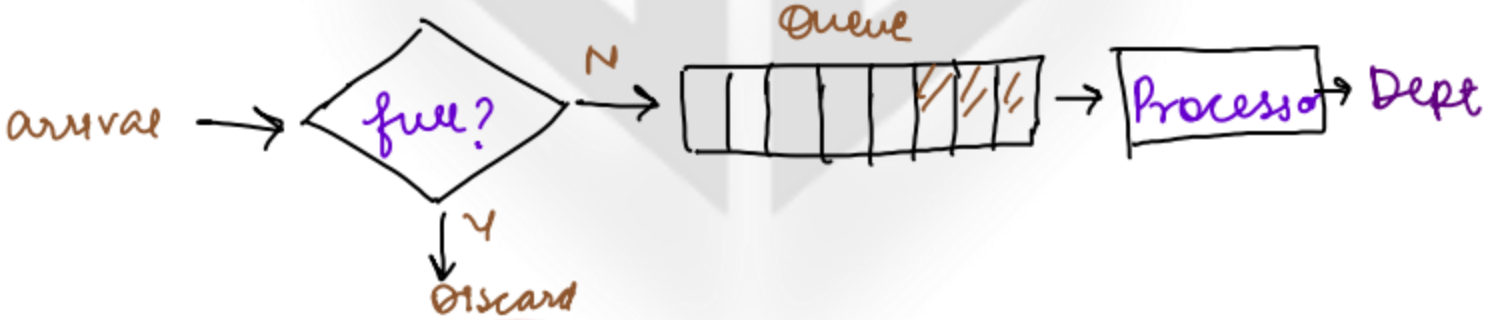
TECHNIQUES TO IMPROVE QoS

- Scheduling
- Traffic Shaping
- Resource Reservation
- Admission Control

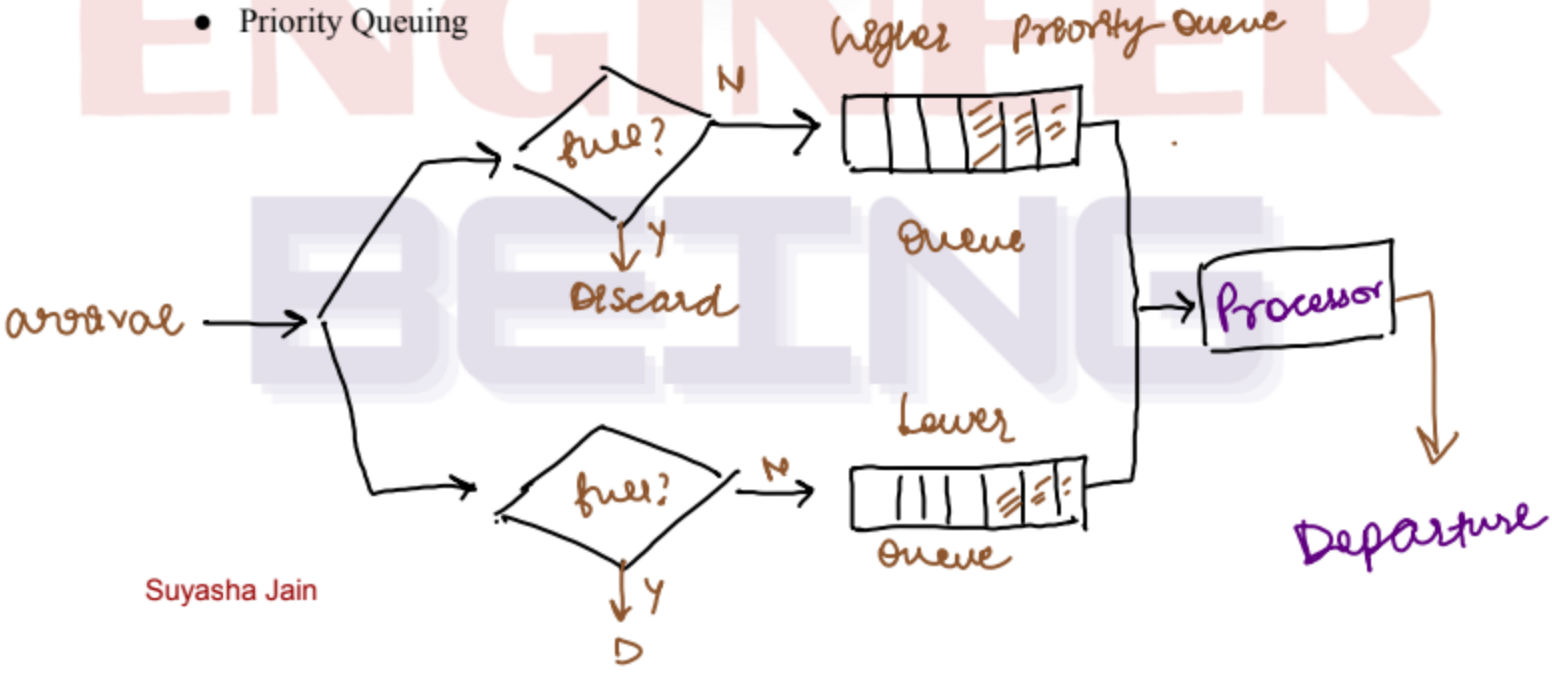
1) Scheduling

Packets from different flows arrive at a switch or router for processing. A good scheduling technique treats the different flows in a fair and appropriate manner.

- FIFO Queuing

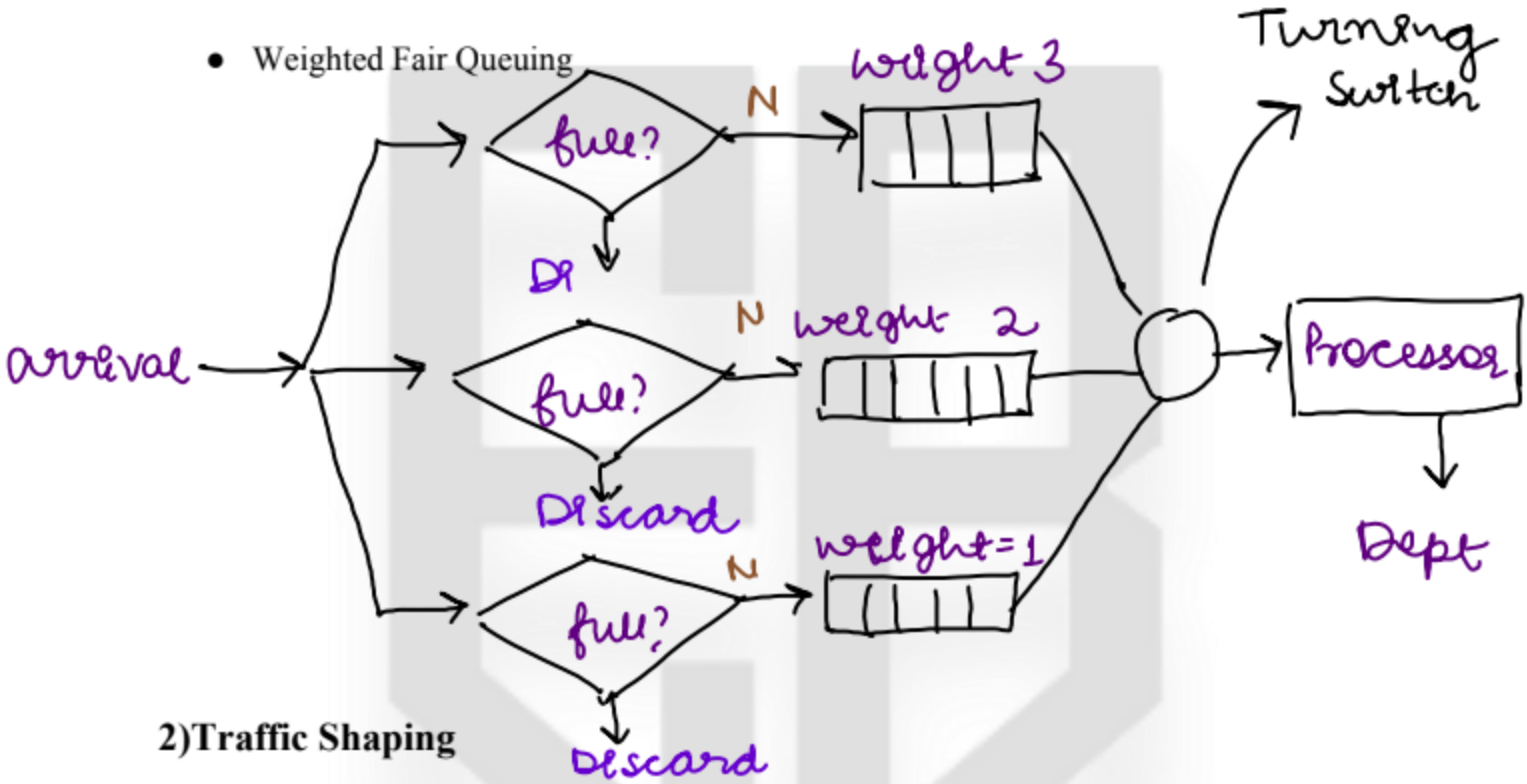


- Priority Queuing



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- Weighted Fair Queuing

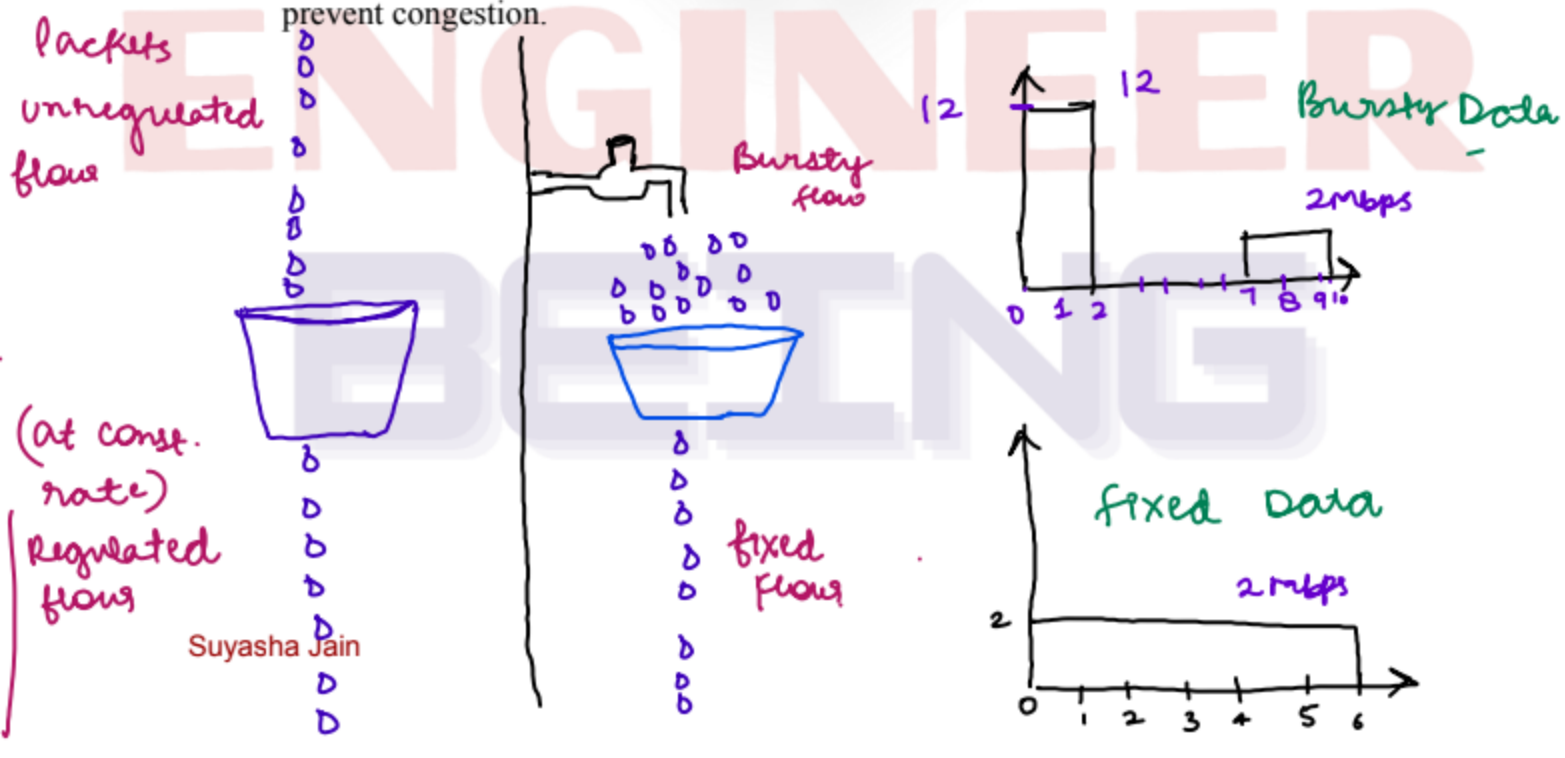


2)Traffic Shaping

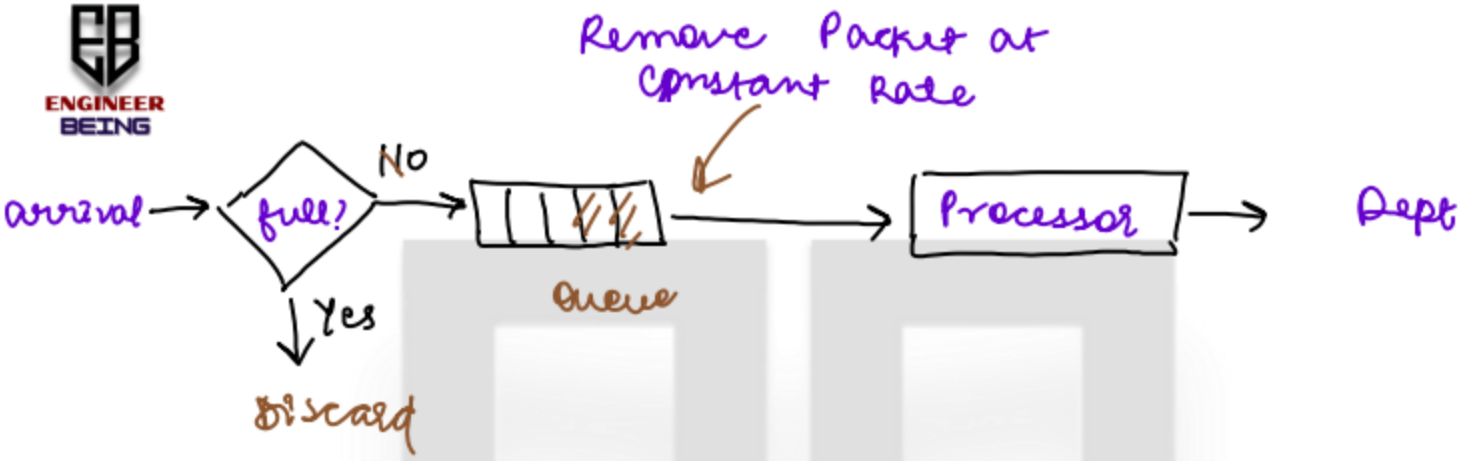
Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network.

Leaky Bucket

- The input rate can vary, but the output rate remains constant.
- The leaky bucket smooths the traffic by sending out data at a uniform rate. Without the leaky bucket, the beginning burst may have hurt the network by consuming more bandwidth than is set aside for this host. We can also see that the leaky bucket may prevent congestion.

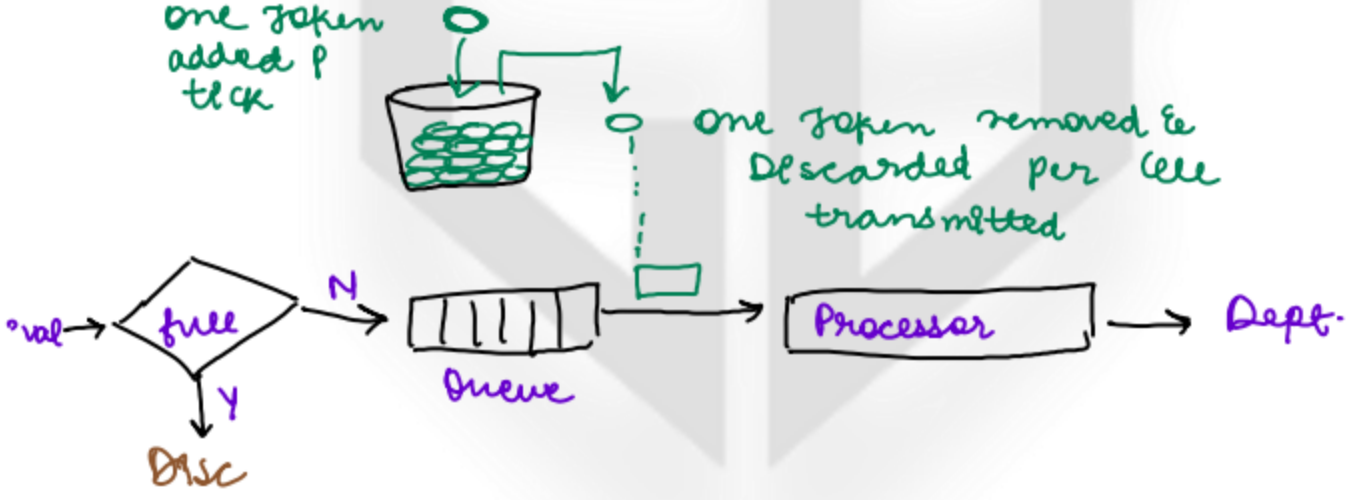


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Token Bucket

The token bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens. For each tick of the clock, the system sends n tokens to the bucket. The system removes one token for every cell (or byte) of data sent.

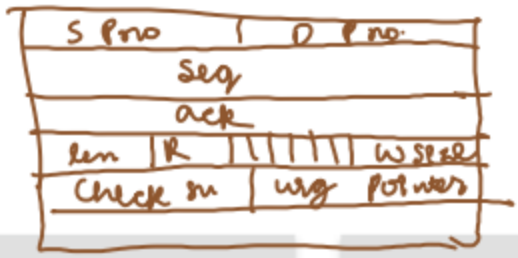


3)Resource Reservation

- A flow of data needs resources such as a buffer, bandwidth, CPU time, and so on.
- The quality of service is improved if these resources are reserved beforehand.

4)Admission Control

- Admission control refers to the mechanism used by a router, or a switch, to accept or reject a flow based on predefined parameters called flow specifications.
- a router accepts a flow for processing, it checks the flow specifications to see if its capacity (in terms of bandwidth, buffer size, CPU speed, etc.) and its previous commitments to other flows can handle the new flow



4 Bits → 1000 = 8 × 4 Bytes = 32 Bytes (20–60)

PYQ

1) The following is the dump of a TCP header in hexa-decimal format: 10M 2018-2019

05320017 00000001 00000000 500207FF 00000000

(i) What is the sequence number?

(ii) What is the destination port number?

(iii) What is the acknowledgement number?

(iv) What is the window size?

① Source & Dest. Port Add - 2 Bytes
②
③ Seq & ack no. - 4 Bytes
④
⑤ Hlen = 1 word ⑥ Reserved = 6 Bits
⑦ Control = 6 Bits ⑧ Window size = 2 Bytes
⑨ Check sum = 2 Byte ⑩ Urg Pointer = 2 Bytes

00000101 — 1 Bytes

05 32 | 00 17 | 00 00 00 01 | 00 00 00 00 | 50 02 | 07 FF | 00 00 | 00 00

1 Bytes

50 → 0101 0000 (Hlen | Reserved)

02 → 0000 0010 (Control)

Each hex Digit = 4 Bits

1. Source Port add = 2 Bytes = 0532
2. Dest Port add = 2 Bytes = 0017
3. Seq no. = 4 Bytes = 00 00 00 01
4. ack no. = 4 " = 00 00 00 00

Hlen = 1 word = 4 Bits = 5

Reserved = 6 Bits 000000

Control = 6 Bits, 0 2 Bits from 0 → 2 Bits of 0 & 2 (00 0010)

Window size = 2 Bytes 07 FF

Check sum = 2 Bytes 00 00

Urgent pointer = 2 Bytes 00 00

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OR

What do you understand about Quality-of-service parameters? List various Quality of service parameters.

2) Explain about the TCP header and working of TCP protocol and differentiate between TCP and UDP with frame format. 10M 2018-19

3) What is the difference between network layer delivery and transport layer delivery? 2M 2018-19

4) Write a detailed note on "TCP vs UDP". 10M 2021-22

5) List out the prime three functions of the transport layer. 2M 2021-22

6) Explain the use of the RST flag in the TCP header. 2M 2021-22

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TCP header itself is of 10 feilds as below and size may vary between 20 to 60bytes

1.Source port - 2 bytes
2.destination port - 2 bytes
3.SEQ NUM-4 bytes
4.ACK NUM- 4 bytes
5.HLEN-1 word
6.RESERVED-6bits
7.CONTROL-6bits
8.WINDOW SIZE-2 bytes
9.CHECKSUM-2 bytes
10.URGENT POINTERS-2bytes

here's example problem from book Forouzan

TCP header(in hex)=05320017 00000001 000000000 500207FF 00000000
since each hex = 4 bits , we need to first split the above hex as such
05 32 00 17 00 00 00 01 00 00 00 00 50 02 07 FF 00 00 00 00

source port is 2 bytes take 05 32 = 1330
next 2 bytes as destination address 00 17 == 23 (default TCP port)
next 4 bytes as sequence number 00 00 00 01 ==1
next 4 bytes as ack 00 00 00 00 == 0
next 4 bits as HLEN 5 ==5 -- this indicates number of sets of 4 bytes which makes the header lenght = 20bytes..
next 6 bits are reserved i.e.0 =0000and 2 bits from hex 0
next 6 bits are control bits = remaining 2 bits from hex 0 and 4 bits of 2
next 2 bytes indicate the window length 07 FF == 2047 bytes
Checksum 2 bytes 00 00 = 0
Urgent pointer 2bytes 00 00 =0

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